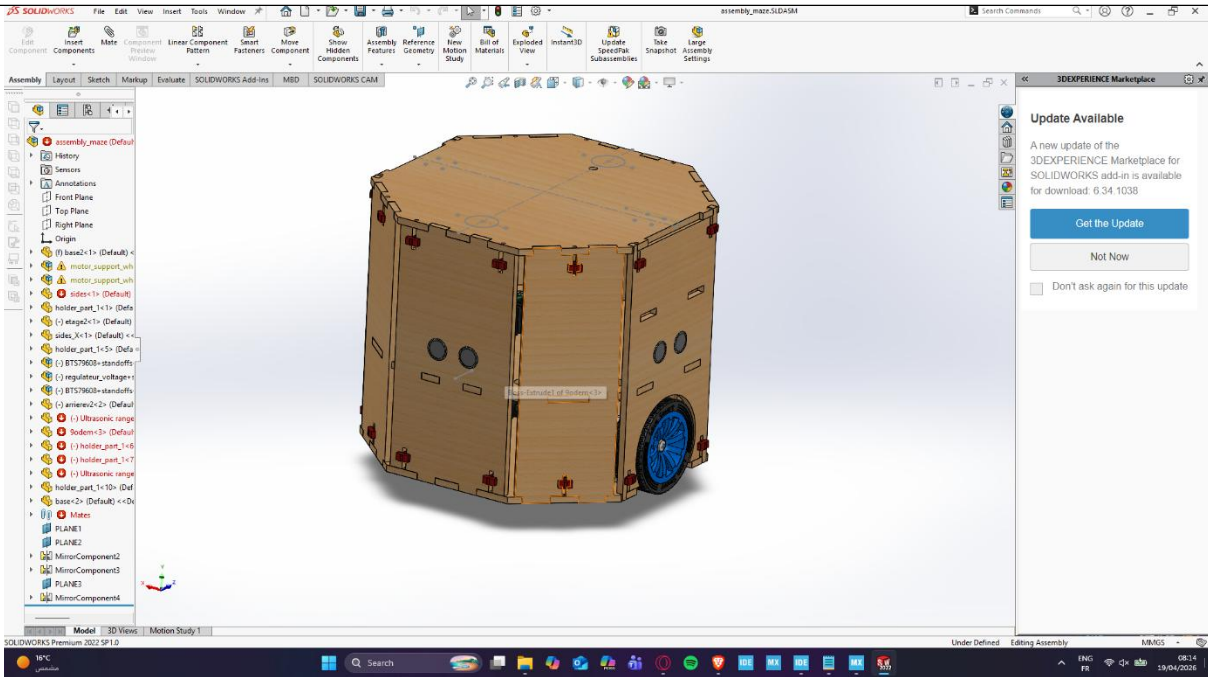
Maze runner robot

1-Mechanical Design Overview :

The mechanical design of the robot was executed using **SolidWorks**, focusing on a robust, modular, and manufacturable structure. We adopted a **decagonal (10-sided) prismatic geometry**, which provides a balanced internal volume for electronics while offering multiple flat surfaces for sensor integration. To ensure a high strength-to-weight ratio and ease of assembly, the chassis was designed using **laser-cut panels**—likely MDF or acrylic—featuring a precise **tab-and-slot (finger joint)** interlocking system. This method ensures self-alignment during construction and increases the structural rigidity of the vertical walls.

From a functional standpoint, the robot utilizes a **differential drive system**, visible through the integrated high-traction wheels mounted on a central axis. The design includes dedicated cutouts for **ultrasonic sensors** on multiple faces to facilitate 360-degree environmental mapping and obstacle avoidance. Internally, the assembly incorporates a multi-tiered layout, using **standoffs and mounting plates** to organize the microcontrollers, motor drivers (such as the BTS7960B visible in your feature tree), and power regulation modules. By utilizing SolidWorks' assembly environment, we were able to perform interference checks and ensure that all mechanical clearances for the motors and wiring were optimized before moving to the fabrication phase.



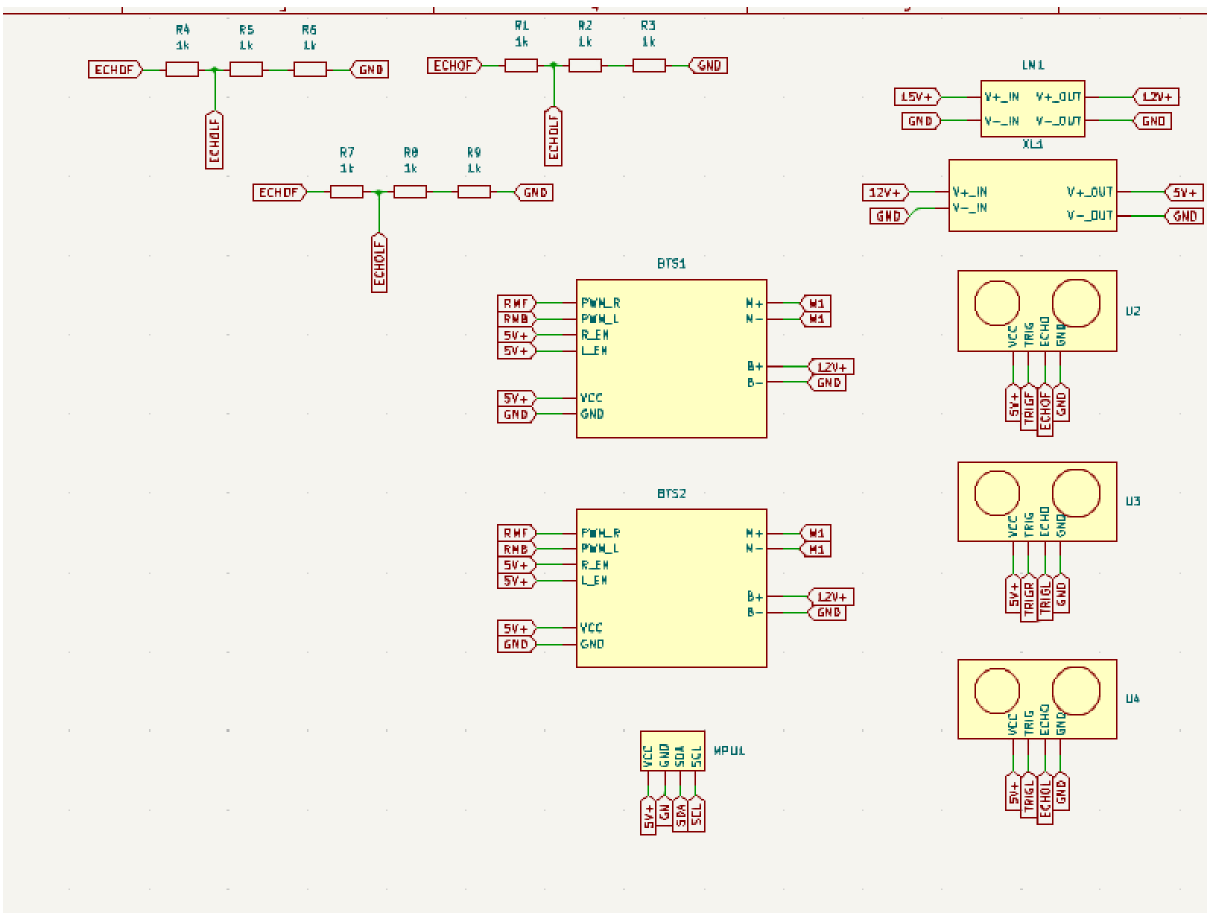
2- The electrical architecture

The electrical architecture of the robot is centered around the **STM32F446RET6** microcontroller, leveraging its high-performance ARM Cortex-M4 core to manage real-time sensor processing and motor control. The system utilizes a dual-power rail design, featuring voltage regulators that step down a **12V source** to a stabilized **5V supply** for the logic gates and peripherals. For mobility, the robot employs two **BTS7960 high-current motor drivers**, which receive PWM signals from the STM32 to provide precise differential steering.

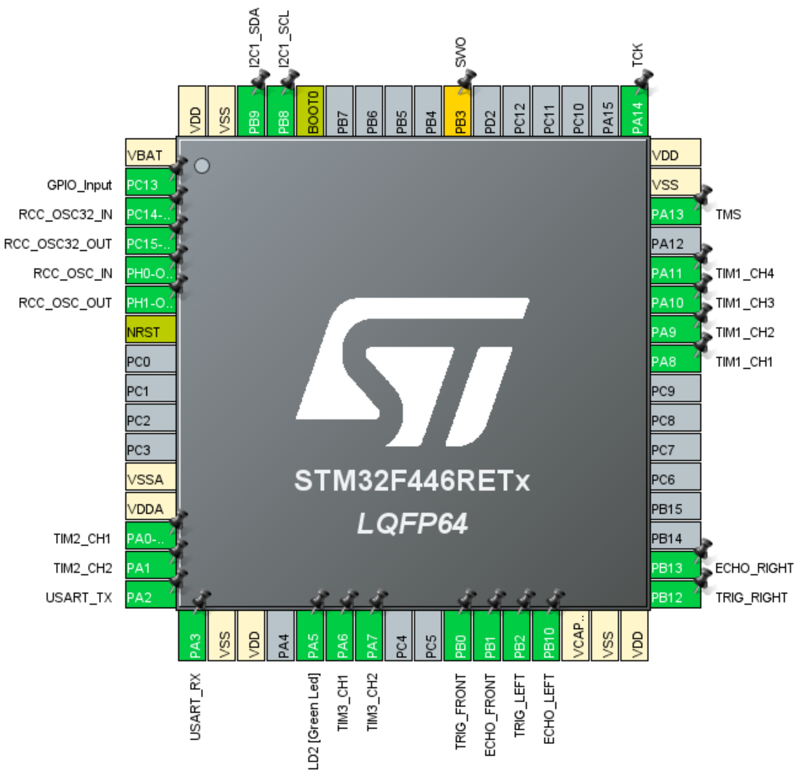
To interface with the environment, the design incorporates three **HC-SR04 ultrasonic sensors** configured for obstacle detection and spatial mapping. A critical feature of this interface is the custom-designed **voltage divider circuits** (using 1kΩ resistor networks), which safely level-shift the 5V Echo signals from the sensors down to the 3.3V logic level required by the STM32 GPIOs. Additionally, an **MPU-6050 IMU** is integrated via the **I2C bus** (pins PB8 and PB9) to provide inertial data for stability and orientation tracking. This integrated approach ensures a high degree of responsiveness and electrical safety, isolating high-power motor noise from the sensitive digital control logic.

Electrical Component Summary

System	Components	Key Interface
Control	STM32F446RETx	PWM, GPIO, I2C
Actuation	2x BTS7960 Drivers	High-Current H-Bridge
Sensing	3x HC-SR04 + MPU6050	Ultrasonic & 6-Axis Motion
Protection	Resistor Voltage Dividers	5V to 3.3V Logic Level Shifting



R4 1k
R5 1k
R6 1k
ECHOF
GND
ECHOLF
R1 1k
R2 1k
R3 1k
R7 1k
R8 1k
R9 1k
LN1
V+_IN
V-_IN
V+_OUT
V-_OUT
15V+
12V+
5V+
XL1
BTS1
BTS2
PWM_R
PWM_L
R_EN
L_EN
VCC
M+
M-
B+
B-
RMF
RMB
M1
U2
U3
U4
TRIG
ECHO
MPU1
SDA
SCL



STM32F446RETx
LQFP64
I2C1_SDA
I2C1_SCL
SWO
TCK
GPIO_Input
RCC_OSC32_IN
RCC_OSC32_OUT
RCC_OSC_IN
RCC_OSC_OUT
TIM2_CH1
TIM2_CH2
USART_TX
TMS
TIM1_CH4
TIM1_CH3
TIM1_CH2
TIM1_CH1
ECHO_RIGHT
TRIG_RIGHT
USART_RX
LD2 [Green Led]
TIM3_CH1
TIM3_CH2
TRIG_FRONT
ECHO_FRONT
TRIG_LEFT
ECHO_LEFT